

GAUTAM SINGH

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EDUCATION

Conestoga College

Guelph, ON

Bachelor of Computer Science (Honors)

Sep 2020 - Dec 2025

Relevant coursework: Machine Learning, Data Structures & Algorithms, Database Design, Cloud Computing (Azure, Databricks), Software Engineering, Object-Oriented Programming

Google Advanced Data Analytics Professional Certificate

Jun 2026

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, C++, Rust

Web & Frameworks: React, FastAPI, Django, REST APIs, Flutter, Streamlit

Data & Cloud: Azure, AWS, Databricks, PySpark, Docker, CI/CD, dbt, Prefect, PostgreSQL, MongoDB, Redis

ML & Tools: PyTorch, scikit-learn, Pandas, NumPy, OpenCV, YOLO, LangChain, Git, ServiceNow, Power BI

EXPERIENCE

Machine Learning Intern

August 2026 - Present

Sphere Global (automated vehicle damage inspection)

Remote

- Contributed to the synthetic training-data pipeline for a 23-camera vehicle damage-detection system.
- Engineered a procedural generator that produces ~12,000 labeled synthetic images, each with a pixel-exact segmentation mask and YOLO label, generated locally.
- Validated the procedural pipeline over a diffusion approach by benchmarking SDXL and FLUX, showing it reproduced damage more accurately, ran significantly faster, and cost less to run.
- Implemented a smarter validation layer that catches and re-segments the 39% of masks capturing the background instead of the vehicle.

ServiceNow Developer & QA Specialist

Sep 2024 - Dec 2024, Sep 2025 - Dec 2025

PAMT Consulting (now Ateko, a Bell Canada company)

Remote

- Reduced onboarding handling time by roughly 30% with an automated workflow built in ServiceNow App Engine Studio and Flow Designer. Staff had processed each new hire by hand, which was slow and error prone.
- Designed and executed test cases across development, test, and production environments, confirming data accuracy and integration reliability before each release and catching defects before end users hit them.

PROJECTS

Cairn

[Live | GitHub](#) /2026

Rust, AWS Lambda, DynamoDB, CloudFront, API Gateway, Terraform

- Built the ingest path as a Rust binary on ARM64 AWS Lambda that responds in 5.1 ms warm and 184 ms cold, doing one DynamoDB write per event and deferring every dimension derivation.
- Deployed the full stack of Lambda, DynamoDB, CloudFront, and API Gateway as 41 Terraform managed resources running live across three sites at zero steady-state cost and made the service cookie-free by hashing visitor identifiers under a key that rotates daily.

Fitmit

[Live | GitHub](#) /2026

Python, FastAPI, React, TypeScript, Redis, Docker, Ghostscript, Pillow

- Cut worst-case compression attempts from twelve to four with a binary search over the quality ladder and shipped the service live on Render and Vercel.
- Extended the compressor to five image formats with no change to the existing search or job queue by abstracting format specific work behind one interface.

The Almanac (Capstone)

[GitHub](#) /2025

Python, Azure Data Factory, Databricks, PySpark, Azure SQL, PyTorch, Django REST, Flutter

- Increased data load performance by ~35% by redesigning the storage layer of the pipeline behind a stock tracking web and mobile application that my team shipped over eight months. The data pipeline ingests, cleans, and processes market data without manual steps.

OPEN SOURCE

actualbudget/actual

[PR #8603](#) /Merged 2026

- Contributed a merged bug fix to a widely used open-source personal finance app, returning a proper error when a password change is attempted with no password method set.